AMARBUYANTAYN, Mongolia

WMO: **443290**

Lat: **44.617N** Lon: **98.700E** Elev: **2103** StdP: **78.48** Time Zone: **8.00 (E08)** Period: **01-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-26.9	-24.7	-32.0	0.2	-23.1	-30.0	0.3	-21.8	12.5	-18.0	11.9	-16.0	2.7	320	0.754	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	9.5	25.8	13.8	24.4	13.0	23.0	12.3	15.9	22.4	14.7	21.6	13.7	20.7	4.6	230	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	13.7	12.7	18.7	12.2	11.4	18.1	10.6	10.3	17.1	52.8	22.7	48.9	21.8	45.6	20.7	22.5	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	13.6	11.8	10.1	DB	-29.5	28.6	2.1	1.5	-31.0	29.6	-32.2	30.5	-33.4	31.4	-34.9	32.5	(4)
(5)				WB	-29.6	17.0	1.9	2.8	-30.9	19.0	-32.1	20.6	-33.2	22.1	-34.6	24.1	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	1.3	-15.4	-13.1	-5.6	2.9	8.6	14.7	17.4	15.9	9.3	1.1	-7.4	-13.3	(6)
(7)			DBStd	12.38	4.81	5.25	5.74	5.32	4.56	3.53	2.97	3.18	4.13	4.62	4.80	4.69	(7)
(8)			HDD10.0	3801	787	646	483	216	80	5	0	2	61	276	522	723	(8)
(9)			HDD18.3	6248	1045	879	741	462	303	118	53	86	272	534	772	982	(9)
(10)			CDD10.0	641	0	0	0	3	36	146	230	186	39	1	0	0	(10)
(11)			CDD18.3	46	0	0	0	0	1	8	24	12	0	0	0	0	(11)
(12)			CDH23.3	253	0	0	0	0	2	54	146	51	0	0	0	0	(12)
(13)			CDH26.7	23	0	0	0	0	0	4	16	3	0	0	0	0	(13)
(14)	Wind		WSAvg	4.7	4.1	4.6	5.1	5.3	5.5	4.5	4.0	4.0	4.5	4.9	5.0	4.5	(14)
(15)	Precipitation		PrecAvg	90	2	1	4	4	6	14	25	17	10	3	2	2	(15)
(16)			PrecMax	140	17	4	23	13	28	63	71	45	36	13	8	8	(16)
(17)			PrecMin	42	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)			PrecStd	25	3	1	5	3	7	15	18	12	8	3	2	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		-1.5	1.6	9.1	17.8	22.7	27.2	28.6	26.7	21.4	14.1	6.2	-0.7	(19)
(20)			MCWB		-4.8	-2.7	1.5	8.5	11.2	14.7	15.6	14.2	10.3	5.5	0.6	-4.9	(20)
(21)		2%	DB		-4.3	-1.0	7.2	15.1	20.4	24.8	26.5	24.6	19.3	11.9	3.6	-3.2	(21)
(22)			MCWB		-7.2	-4.8	0.3	6.1	10.1	13.0	14.1	13.4	9.6	4.7	-0.9	-6.4	(22)
(23)		5%	DB		-6.6	-3.3	5.2	13.2	18.4	23.2	24.9	23.4	17.6	10.1	1.4	-5.0	(23)
(24)			MCWB		-8.9	-6.4	-0.8	4.8	8.8	12.1	13.2	12.6	8.8	3.4	-2.4	-7.8	(24)
(25)		10%	DB		-8.6	-5.3	3.1	11.3	16.5	21.5	23.4	22.1	16.2	8.2	-0.4	-6.9	(25)
(26)			MCWB		-10.5	-7.9	-2.1	3.8	7.8	11.3	12.6	12.0	8.2	2.2	-3.7	-9.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		-4.2	-2.4	2.3	10.2	13.8	16.5	18.2	17.2	12.2	7.6	1.3	-4.4	(27)
(28)			MCDB		-1.8	1.0	8.2	16.7	20.7	22.7	23.5	24.1	18.4	11.7	5.6	-1.2	(28)
(29)		2%	WB		-6.9	-4.8	0.6	7.2	11.5	14.6	16.3	15.6	10.8	5.4	-0.6	-6.4	(29)
(30)			MCDB		-4.7	-1.4	6.3	13.6	18.4	22.2	22.6	22.4	16.9	10.2	3.1	-3.4	(30)
(31)		5%	WB		-8.9	-6.3	-0.7	5.5	9.9	13.4	15.0	14.3	9.7	3.9	-2.2	-7.7	(31)
(32)			MCDB		-6.8	-3.5	4.8	12.3	16.5	20.8	21.9	20.8	16.0	9.5	1.2	-5.2	(32)
(33)		10%	WB		-10.4	-7.8	-1.9	4.1	8.5	12.3	14.0	13.1	8.7	2.6	-3.7	-9.1	(33)
(34)			MCDB		-8.6	-5.4	2.9	10.6	15.3	19.6	20.8	20.2	15.0	7.7	-0.7	-7.0	(34)
(35)	Mean Daily Temperature Range		MDBR		8.8	9.5	10.3	10.5	10.5	10.1	9.5	9.5	9.5	9.4	8.3	8.3	(35)
(36)		5% DB	MCDBR		10.1	10.8	11.5	12.0	11.9	11.4	10.9	10.2	10.2	10.4	9.5	9.9	(36)
(37)			MCWBR		8.1	8.2	7.7	7.6	7.2	6.1	5.4	5.0	5.9	6.6	7.0	7.8	(37)
(38)		5% WB	MCDBR		9.7	10.7	11.1	11.4	11.1	10.3	9.7	9.2	9.5	10.0	9.2	9.8	(38)
(39)			MCWBR		7.9	8.2	7.7	7.7	7.6	6.4	5.3	5.2	5.9	7.0	6.9	7.8	(39)
(40)	Clear-Sky Solar Irradiance		taub		0.238	0.267	0.309	0.346	0.357	0.378	0.371	0.349	0.317	0.283	0.253	0.236	(40)
(41)			taud		2.397	2.324	2.209	2.159	2.186	2.202	2.266	2.315	2.364	2.400	2.410	2.394	(41)
(42)			Ebn at Noon		907	929	923	912	910	889	891	900	906	898	876	872	(42)
(43)			Edh at Noon		72	94	122	141	142	141	131	120	104	86	71	65	(43)
(44)	All-Sky Solar Radiation		RadAvg		2.15	3.26	4.86	6.26	7.28	7.13	6.78	6.12	5.30	3.87	2.51	1.90	(44)
(45)			RadStd		0.14	0.21	0.27	0.25	0.38	0.42	0.28	0.34	0.22	0.13	0.12	0.15	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page